

CompSci 234/NetSys 210
Advanced Topics in Networking
Winter 2019

Unit 16: Quantum Internet

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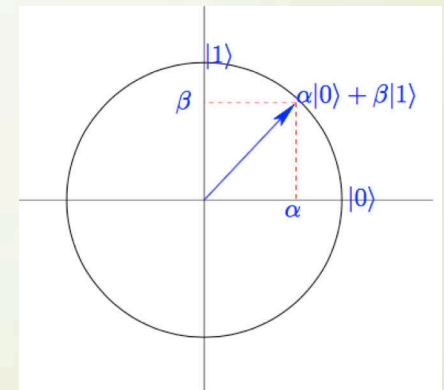
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slides adopted from IRTF QIRG Bangkok Meeting, Nov. 2018

Quantum Bits

- Quantum mechanics: there exist **superpositions** of 0 and 1 states.
- Mathematically, the state of a quantum bit (qubit) is described by:
 - $\alpha|0\rangle + \beta|1\rangle$
- where α and β are complex number, and $|\alpha|^2 + |\beta|^2 = 1$
- Measuring the above state gives $|0\rangle$ with prob. $|\alpha|^2$ and $|1\rangle$ with prob. $|\beta|^2$
- Then the qubit gets into the **measured state**

Note: $|\phi\rangle$ is called Dirac notation, but it's nothing but a column vector



Vision of Quantum Internet

- ▶ Enabling quantum communications among (quantum) computers anywhere on earth.



Why Construct a Quantum Internet?

For quantum communications

- Quantum secure communications
 - Secure Identification
 - Clock synchronization
 - Protocols for distributed systems
 - Combining telescopes
 - Testing Physics
 - Exponential savings in communications
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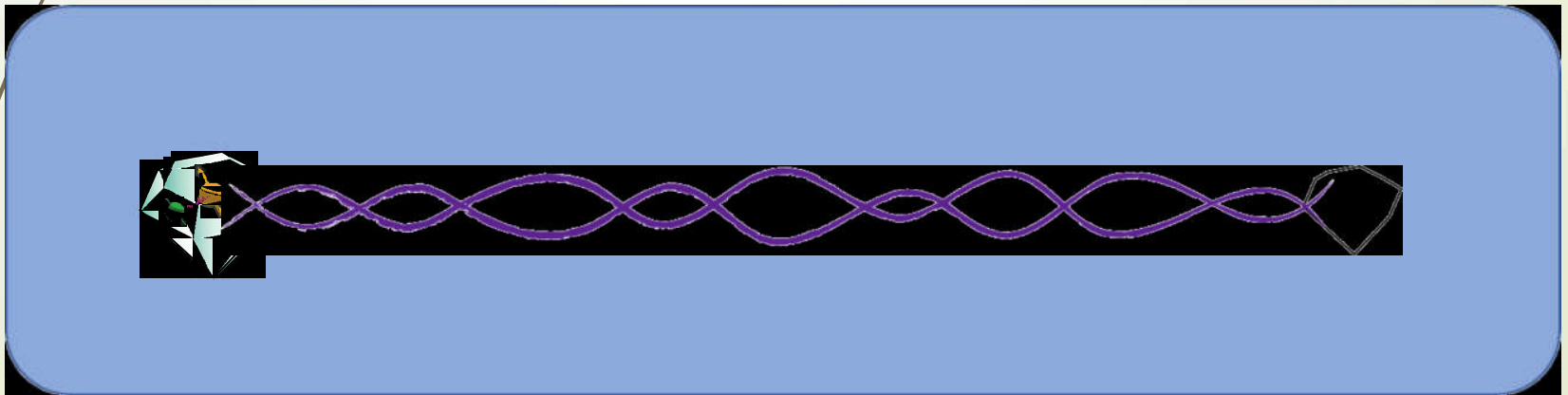
For quantum computations

- Linking small quantum computers
- Access the quantum “mainframe”



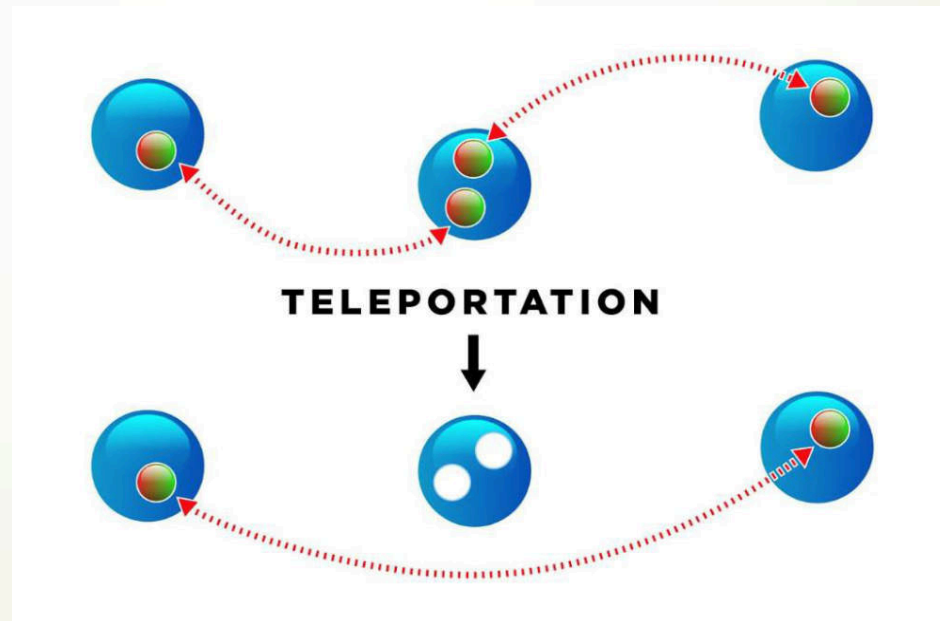
Entanglement

- Properties of entanglement:
 - **Complete coordination:** Measurement outcomes are random but perfectly correlated.
 - **Inherently private:** No one can have any share of the entanglement.

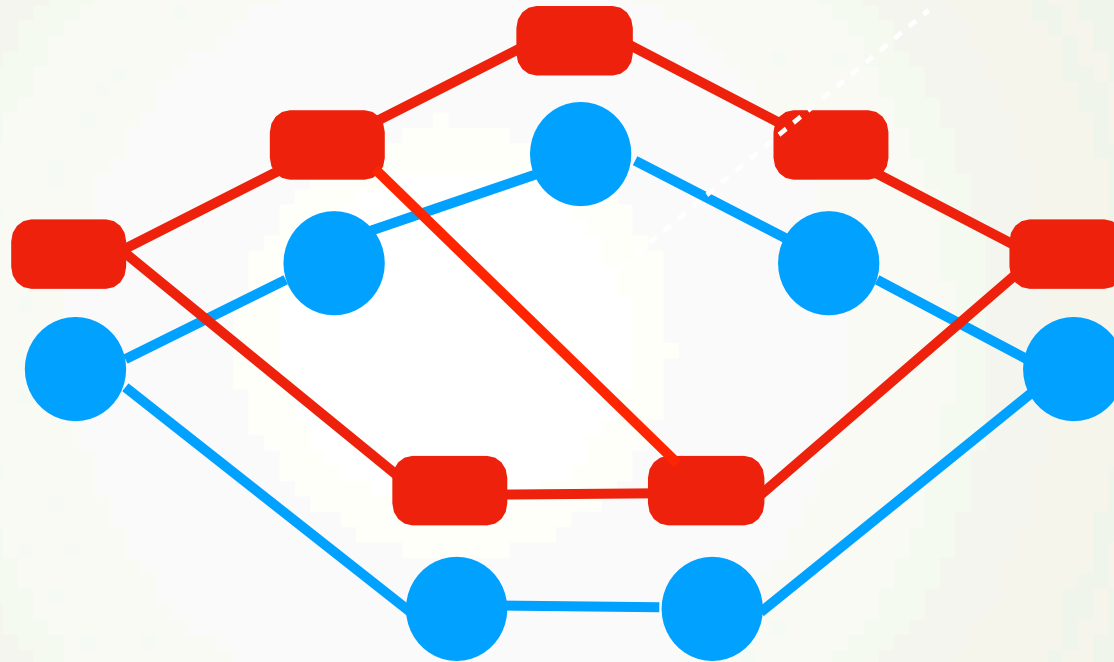


Quantum Repeater

- Entanglements can only be created in short distance
→ need teleportation or entanglement swapping



Quantum Internet (with Classical Internet)

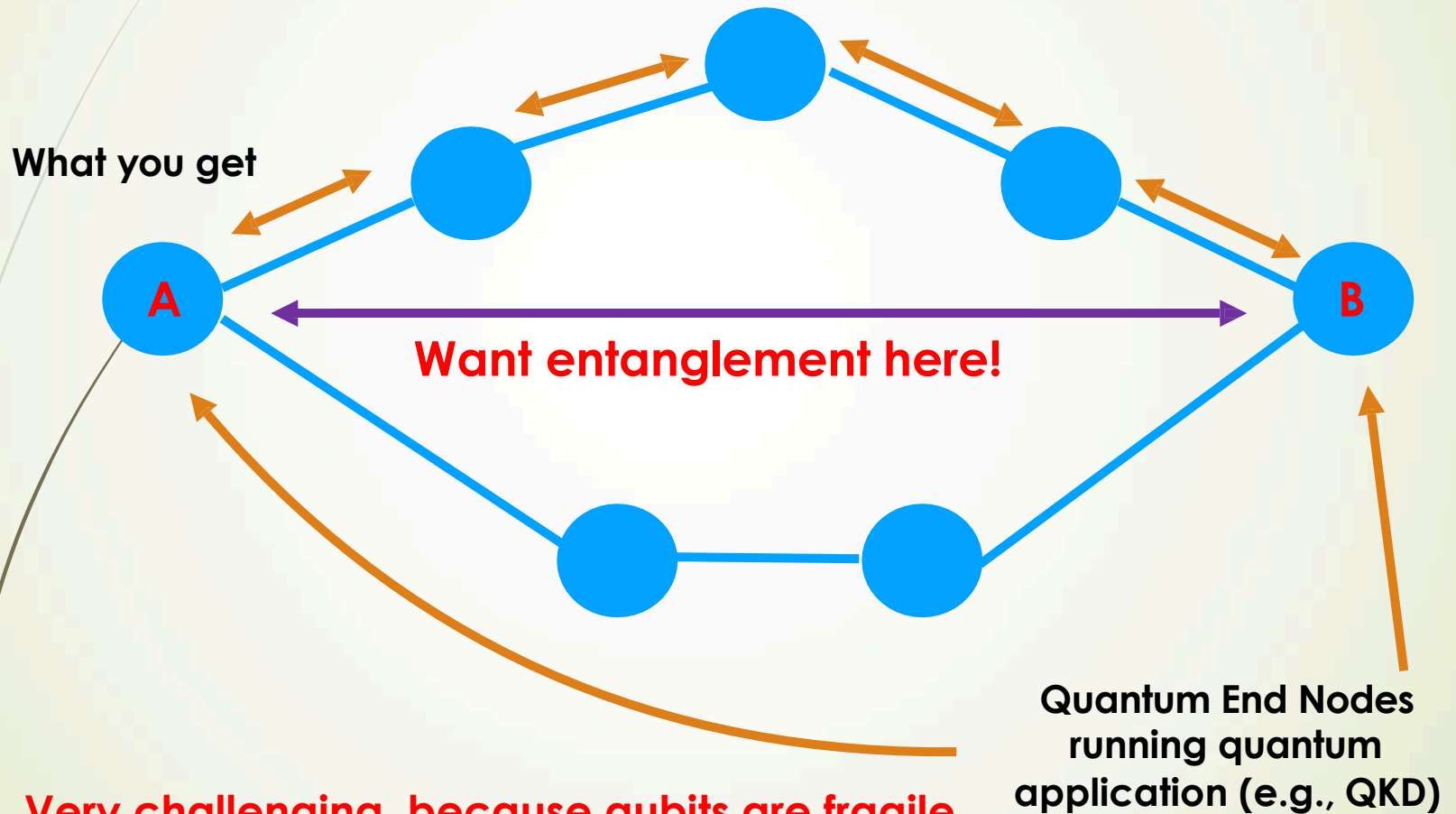


-  Quantum node
-  Quantum link
-  Classical node
-  Classical link

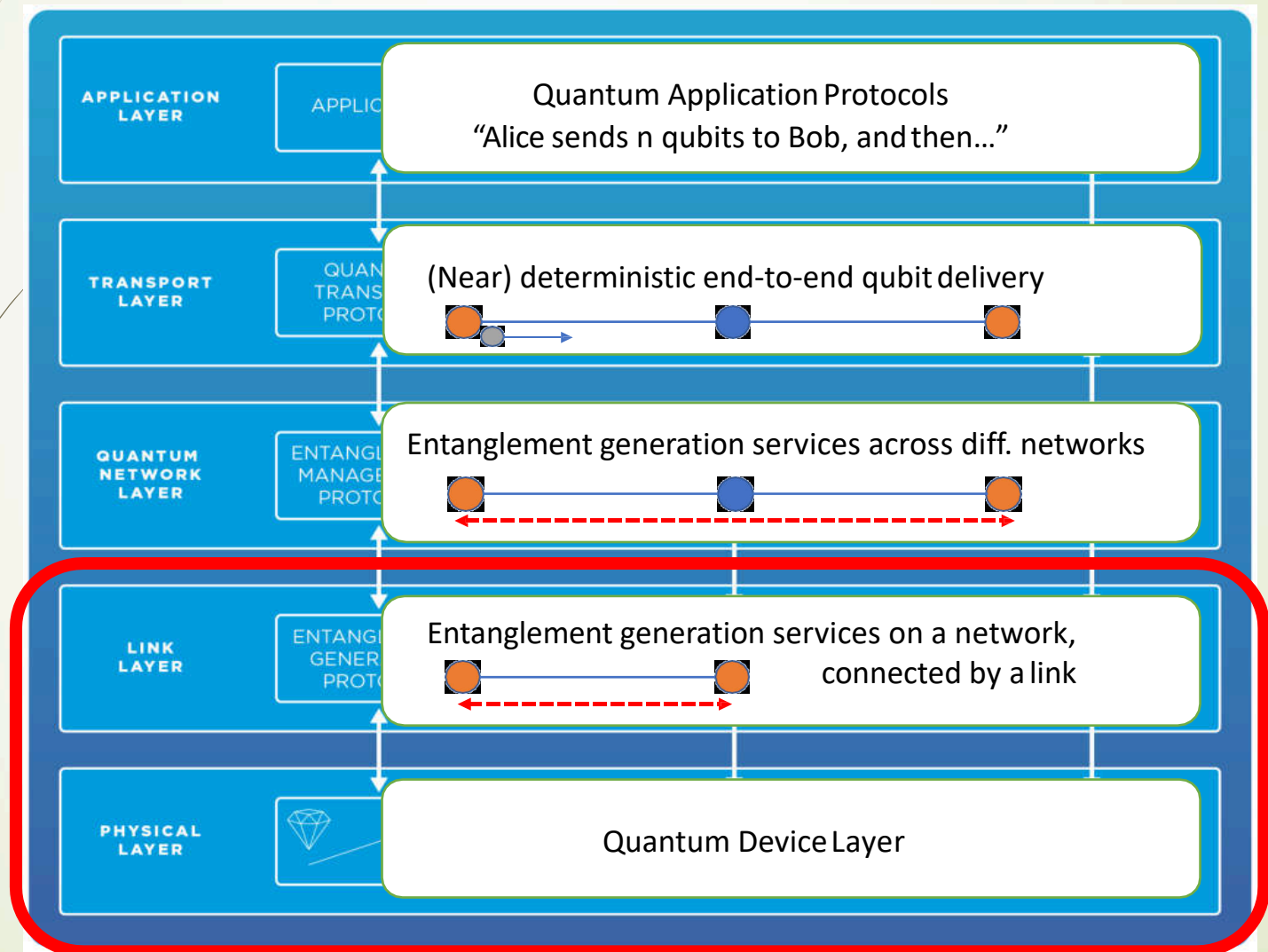
Classical network parallels the quantum network

Classical nodes initiate quantum operations on the quantum nodes

Key Objective of Quantum Internet



Layering Structure (Delft Version)



Six Stages Proposed in [WKH'08]

